Experiment No. 13: Case Study on Low-Code Limitations

1. AIM

To study the limitations of low-code development platforms and understand their impact on application development using Zoho Creator.

2. THEORY

Low-code development platforms allow users to build applications with minimal hand coding using visual tools, drag-and-drop components, and predefined workflows. However, they have certain limitations.

Major limitations include:

- **Limited Customization:** Complex features may require additional coding or may not be supported.
- **Platform Dependency:** Applications depend on the platform's tools, features, and pricing.
- **Performance Issues:** Applications handling large amounts of data or complex operations may face performance challenges.
- **Integration Challenges:** Connecting certain external systems may require APIs or custom code.
- **Security Concerns:** Incorrect permission settings can expose sensitive information.
- **Scalability Limitations:** Some applications may require architectural changes as user numbers and data volumes increase.
- **Vendor Lock-in:** Migrating applications to another platform may require significant effort.

3. PROCEDURE

1. Select a sample application, such as a Vehicle Service Management system, built using Zoho Creator.
2. Identify the application's requirements and features.
3. Examine the forms, reports, workflows, and integrations used.
4. Identify limitations in customization, performance, and integration.
5. Analyze security, scalability, and platform dependency concerns.
6. Suggest suitable improvements, such as optimized workflows, APIs, and access controls.
7. Evaluate whether the low-code platform meets the application's requirements.

4. CASE STUDY

Application: Vehicle Service Management System

Zoho Creator can be used to build forms for service requests, reports for repair tracking, and workflows for status updates. However, highly customized interfaces, complex integrations, and advanced processing may require Deluge scripting or external services. As the application grows, performance, security, and migration requirements must also be considered.

Possible solutions:

- Use optimized Deluge scripts and reports.
- Integrate external services through APIs.
- Configure appropriate user permissions.
- Test application performance regularly.
- Evaluate migration requirements before committing to the platform.

5. RESULT

The limitations of low-code development platforms were studied through a case study of a Vehicle Service Management System. The study showed that low-code platforms accelerate development but may require custom coding, integrations, and careful planning to address complex requirements.